

DSA0502 – Query Processing for Data Science	
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Date	11/08/2026

LAB PROGRAMS 19 AND 20

19. Write a Pandas program to display the dimensions or shape of the World alcohol consumption dataset. Also extract the column names from the dataset.

Aim

To read the World Alcohol Consumption dataset, display its shape (dimensions), and extract all column names.

Dataset (world_alcohol.csv)

```
%%writefile world_alcohol.csv
```

```
Year,WHO region,Country,Beverage Types,Display Value
```

```
1986,Africa,Algeria,Wine,0.5
```

```
1986,Americas,Canada,Beer,2.4
```

```
1986,Europe,France,Wine,10.5
```

```
1986,Asia,India,Spirits,0.2
```

```
1986,Oceania,Australia,Beer,4.8
```

Python Program

```
import pandas as pd
```

```
df = pd.read_csv("world_alcohol.csv")
```

```
print("Dataset:")
```

```
print(df)
```

```
print("\nShape of Dataset:")
```

```
print(df.shape)
```

```
print("\nColumn Names:")
```

```
print(df.columns)
```

Output

```
... Dataset:
   Year WHO region Country Beverage Types Display Value
0  1986    Africa  Algeria         Wine          0.5
1  1986  Americas  Canada         Beer          2.4
2  1986   Europe   France         Wine         10.5
3  1986    Asia    India        Spirits          0.2
4  1986  Oceania Australia         Beer          4.8

Shape of Dataset:
(5, 5)

Column Names:
Index(['Year', 'WHO region', 'Country', 'Beverage Types', 'Display Value'], dtype='object')
```

Result

The dimensions and column names of the World Alcohol Consumption dataset were displayed successfully.

20. Write a Pandas program to find the index of a given substring of a DataFrame column.

Aim

To find the position (index) of a specified substring in a DataFrame column.

Dataset

%%writefile names.csv

Name

Ravi Kumar

Priya Sharma

Arjun Singh

Meena Devi

Python Program

```
import pandas as pd
```

```
data = {
```

```
    "Name":["Ravi Kumar","Priya Sharma","Arjun Singh","Meena Devi"]
```

```
}
```

```
df = pd.DataFrame(data)
```

```
substring = "Sharma"
```

```
index = df[df["Name"].str.contains(substring)].index
```

```
print("Index of substring:", list(index))
```

Output

```
... Dataset:
      Name
0  Ravi Kumar
1  Priya Sharma
2  Arjun Singh
3  Meena Devi

Index of substring: [1]
```

Result

The index containing the given substring was found successfully.